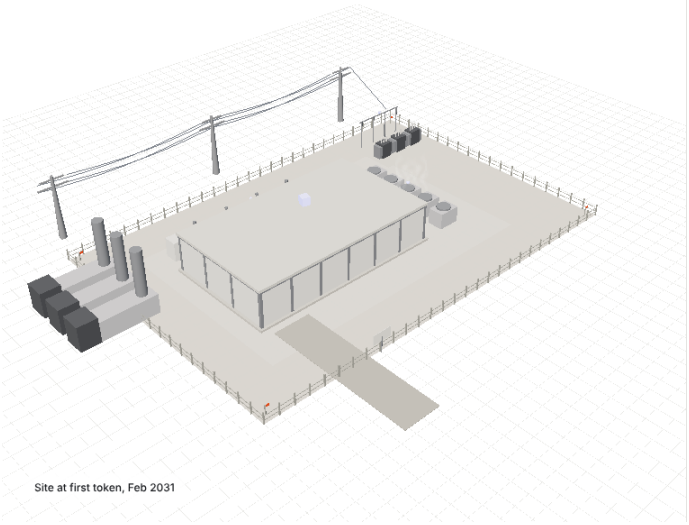
Time to First Token

How many months from bare land to the first token, which choices buy those months back, and where every number came from.



Site at first token, Feb 2031

TIME TO FIRST TOKEN

51 months

First token Feb 2031, full grid power Aug 2032

FACILITY AND POWER	OPEX PER YEAR
$2.9B to $7.4B	$359M to $569M

DESIGN PUE	WATER PER YEAR
1.25	49M gal

300 MW in Virginia on PJM, hybrid power (gas turbines), N+1, liquid cooling, dry coolers, 2,500 racks at 120 kW.

Critical path: Gas turbines, Cx L5, GPU racks installed, Cluster bring-up and burn-in.

Data pack 2026.10.24, 205 values, each with its source. A planning model, not a quote.

github.com/builtbyaarya/time-to-first-token

The problem

Capacity is sold before it is finished: 80.4% of the capacity under construction in North America was already committed in the first half of 2026 (CBRE), so the delivery date is the product. Grid waits of three to seven years are reported for new large loads (Caterpillar, in Data Center Frontier); a large-load request in Texas would carry a study fee of $100,000 to $300,000 and $50,000 per MW of security under the rules the PUC proposed in March 2026 (Foley). A screen that is a year wrong costs real money, and most screens cannot say where their numbers came from.

What the tool does

Set the state, the IT load, the power strategy (grid, on-site bridge, hybrid), redundancy, cooling and accelerator generation; the site builds itself in 3D, stage by stage, and the plan updates live. You get the critical-path schedule, the levers that save or cost months and dollars, capex and opex ranges, PUE and water, a risk register, and seven stress tests (transformer slip, interconnection study delay, permit challenge, turbine lead time, utility energization delay, GPU allocation, labour).

Why the numbers can be checked

Every figure is read from a dated, versioned data pack: 205 values from 77 source pages (EIA, LBNL, EPA, the ISOs, FERC, Dominion, permitting pages, Uptime, Cushman & Wakefield, JLL, NVIDIA, vendor sheets, filings), 133 read straight off the source, 54 derived from them with the arithmetic recorded, 18 labelled assumptions. Before release an independent audit ran 252 checks against the sources and an industry reviewer, in the role of a development director, raised 25 items; the five blockers and the failing checks were fixed and both reports ship in the repo.

Three sample plans

300 MW in Northern Virginia on hybrid power: first token in 51 months. 100 MW in Texas on the grid: 56 months. 1,000 MW in Iowa on bridge power: 62 months.

The study

Enter an email on the page and a feasibility study for your exact inputs arrives as a PDF within a day: executive summary, site and grid, power strategy, permitting path, design basis, schedule, capex and opex, sensitivities, risks, and a source appendix; every figure carries a source number. It is a planning study, not engineering advice and not a quote.

Where I most want critique

The grid energization waits by state, the long-lead equipment times (transformers, switchgear, turbines), and the construction and commissioning durations. If a number looks wrong, the source it came from is one click away.

Try it

Hosted, no sign-up, nothing stored:
`builtbyaarya.github.io/time-to-first-token`
Code, data pack and audit reports:
`github.com/builtbyaarya/time-to-first-token`

Model figures from data pack 2026.10.24. Built by Aarya (builtbyaarya).